25901851

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION April 2026

second SEMESTER

Minor Pathway Courses (MPC) - MG2MPCESM100 - ENVIRONMENTAL EDUCATION AND BIODIVERSITY CONSERVATION

(2025 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **25** questions

Each question carries **1** marks

1. **State anthropogenic environment.** [K] / [CO1]

   An anthropogenic environment includes components that have been introduced by human beings depending on their needs and requirements.

2. **Assess the strategies of environmental education.** [E] / [CO1, CO2]

   (Write any two points)

   1. Cultivating a sense of environmental responsibility among students.

   2. Fostering critical thinking and problem solving in environmental education.

   3.Adopting experiential learning approaches

   4. Collaborating with local communities and organizations.

   5. Using Technology and Digital Tools to Enhance Environmental Education

3. **Explain the term Environmental Studies.** [U] / [CO1, CO2]

   Environmental study is the scientific study of all the components or factors that make or influence our life-supporting biophysical environment.

4. **Recommend the philosophical approaches in environmental ethics.** [E] / [CO2]

The environmental ethics or Dharma of Ecology teaches us how to treat nature. It includes cultures and traditions that have influenced human relationship with nature, the place of man in nature, rights and duties of individuals, the moral standings of humans and duties of future generations.

5. **Appraise Deep ecology.** [E] / [CO1]

Deep Ecology is a radical environmental philosophy introduced by the Norwegian philosopher Arne Naess in 1973. It argues that nature has intrinsic value, meaning that plants, animals and ecosystems have value in themselves, not just because they are useful to humans. According to Deep Ecology, human beings are only one part of the ecosystem and do not have the right to dominate other forms of life.

6. **Recommend any two principles of environmental ethics.** [E] / [CO2]

1. Protection of renewable natural resources.

2. Conservation of non-renewable resources.

7. **Describe the objectives of environmental education.** [U] / [CO1, CO2]

1.Awareness

2. Knowledge

3. Attitudes,

4.Skills,

5.Evaluation ability

6. Participation

8. **Recommend the need of environmental education**. [E] / [CO1, CO2]

Environmental education is needed to create awareness and develop responsible behavior for the protection and conservation of the environment.

9. **Outline one international NGO working in environmental protection in India**. [An] / [CO3]

Greenpeace, WWF, Conservation International, World Resources Institute, Friends of the Earth International etc.

10. **Recommend one woman leader associated with Chipko movement.** [E] / [CO3, CO4]

Gaura Devi

11. **Propose the aim of Appiko movement.** [C] / [CO3, CO4]

Aim of Appiko movement is to stop commercial logging and protect natural forests in the Western Ghats.

12. **Identify the type of forest which was threatened by Silent Valley Hydroelectric Project**. [E] / [CO3, CO4, CO5]

Tropical Rain Forest

13. **Recommend the scope of public awareness.** [E] / [CO3]

Write any one of the following scopes.

1. Developing sensitivity and awareness towards environmental issues.
2. Inculcating and imbibing the stringent need for conservation of natural resources and surrounding environment.
3. Encouraging active participation of the people in environmental protection and development.
4. Developing expertise for the active identification and finding remedy to environmental problems.
5. Assessment of socio-economic, ecological and aesthetic aspects and impacts of various environmental programmes initiated to improve environmental health.

14. **Evaluate environmental management.** [E] / [CO3]

Environmental management is essential for balancing development with environmental protection. It helps in the sustainable use of natural resources, reduces pollution, conserves biodiversity, and ensures long-term ecological stability

15. **Identify the role the tribal communities play in forest conservation.** [E] / [CO3, CO4, CO5]

Tribal communities protect forests through sustainable use of resources and traditional practices that conserve biodiversity.

16. **Select any two environmental movements in India**. [E] / [CO3, CO4]

(Write any one of the movement.)

Bishnoi Movement, Save Silent valley movement, Chipko movement, Appiko movement, Narmada Bachavo Andolan, Jungle Bachavo Andolan, Tehri Dam movement etc.

17. **Recommend the leader of Narmada Bachavo Andolan (NBA).** [E] / [CO3, CO4]

Medha Patkar

18. **Evaluate the need of public awareness.** [E] / [CO3]

Public awareness is crucial for solving environmental problems because it encourages people to adopt eco-friendly behaviors, supports conservation efforts, and ensures participation in environmental protection. It also helps in the effective implementation of laws and policies.

19. **Recommend the name of plant which yield drug for Cancer.** [E] / [CO3, CO4]

Taxol an anticancer drug obtained from the bark of yew tree.

20. **Evaluate on Man- Wildlife conflicts**. [E] / [CO3, CO4]

Man–wildlife conflict is a growing environmental issue caused by habitat loss, deforestation, and human encroachment into wildlife areas. It leads to damage to crops, property, and sometimes loss of human and animal lives.

21. **Evaluate the term "biodiversity."** [E] / [CO3]

Biodiversity refers to the variety of living organisms at genetic, species, and ecosystem levels.

22. **Evaluate anthropogenic extinction.** [E] / [CO3, CO4]

It refers to the disappearance of species from the earth due to human activities.

23. **Propose any two important National Parks in India.** [C] / [CO3, CO4, CO5]

Silent valley National Park, Gir National Park, Eravikulam National Park etc

24. **Evaluate on poaching.** [E] / [CO3, CO4]

Killing of prohibited wild animals for illegal trading of wildlife trading is called poaching.

25. **Appraise the three perspectives of community diversity**. [E] / [CO3]

Alpha diversity, Beta diversity and gamma diversity.

26. **Describe botanical gardens.** [U] / [CO3, CO4, CO5]

Botanical gardens are specialized institutions where a wide variety of plants are grown, collected, and maintained for scientific study, conservation, education, and recreation.

27. **Describe the biodiversity hotspots in India.** [U] / [CO3, CO5]

Western Ghats and Eastern Himalaya

[1x25 = 25]

Part B

Short Essay

Answer any 5 questions

Each question carries 5 marks

28. **Interpret the multidisciplinary nature of environmental studies.** [U] / [CO1, CO2]

Environmental Studies is a multidisciplinary programme created to promote

the study of our natural surroundings, designs to minimize the degradation of those surroundings, and the social and political tools needed to preserve and improve the environment.

- The important disciplines related to the subject of Environmental Studies are

  Biological Sciences: Biological sciences form an important component of Environmental Sciences as it is impossible to study the environment without acquiring knowledge of the organisms living in it. For this purpose, it is important to understand the basic concepts of botany, zoology, microbiology, anthropology, biotechnology, etc. to study the different aspects of environment and their inter-relationships.

  Physical Sciences: Problems such as acid rain, ozone-layer depletion, etc. can be understood properly only if one has an understanding of simple laws of chemistry. Similarly, energy transformations and bio-geochemical cycles that are the main processes that comprise the field of ecosystem ecology require knowledge input from physics. In the same way, Environmental Studies is linked to subjects such as biophysics, biochemistry, geophysics, etc.

  Other disciplines: Geography is an area of study relevant to anyone who is concerned with the relationship between society and the environment. Disciplines such as geology, geography, space science, population studies, sociology, remote sensing, etc. combine both the physical and the social sciences and provide skills and conceptual frameworks needed to understand the complex processes of environment. Issues of social justice and of environmental degradation demand that we explore ways to live in harmony with nature. Similarly, anthropology offers vital insights into the importance of cultural difference existing among individuals.

29. **Outline the goals of environmental education programme launched by UNESCO in 1975.** [K] / [CO1]

1. To facilitate the coordination, joint planning, pro-programming activities essential to teach development of an international progamme of environmental education.

2. To promote the international exchange of ideas and information, pertaining to environmental education.

3. To coordinate research to understand better the various phenomena involved in teaching and learning about environment.

4. To formulate and evaluate new methods, materials and programmes (both in school and out-of-school for youths and adults) in environmental education.

5. To train and retrain personnel and staff for the environmental education programmes

6. To prepare advisory services to member-States relating to environmental education.

30. **Propose Jungle Bachavo Andolan**. [C] / [CO3, CO4]

The Jungle Bachao Andolan (Save the Forest Movement) was an important environmental movement in India that aimed to protect natural forests from destruction. The movement began in the early 1980s in the Singhbhum district of present-day Jharkhand, which was then part of Bihar. It was mainly led by tribal communities who depended on forests for their livelihood.

The main reason for the movement was the government's decision to replace natural Sal forests with teak plantations for commercial purposes. Teak is a valuable timber tree, but planting it in place of diverse natural forests disturbed the ecological balance. Natural forests provide fruits, medicinal plants, firewood, fodder, and shelter for wildlife. They are also deeply connected to the culture and traditions of tribal people. The introduction of commercial plantations threatened both biodiversity and the survival of local communities.

The tribal people strongly opposed this decision. They argued that forests are not merely sources of timber but are essential for life and livelihood. Through protests, awareness campaigns, and community action, they demanded the protection of natural forests and recognition of their traditional rights over forest resources.

The Jungle Bachao Andolan highlighted the importance of conserving biodiversity and promoting sustainable development. It also emphasized people's participation in forest management. The movement brought attention to the harmful effects of replacing natural ecosystems with monoculture plantations.

In conclusion, the Jungle Bachao Andolan is a significant environmental movement in India. It demonstrated the strong bond between tribal communities and forests and stressed the need to protect natural resources for present and future generations.

31. **Recommend formal education and non-formal education.** [E] / [CO1, CO2]

Environmental education can be divided into two major categories: formal and non-formal.

1. Formal Environmental Education, It includes teaching and training of students of primary, secondary, higher secondary and university levels as well as teachers and environmental professionals.

i) At primary school stage, students are sensitized about the environment through audio-visual and field visits.

(ii) At lower secondary stage, stress is on real life situations conservation and sustainable development. Teaching strategy includes those used at primary level along with general science, teaching, practicals and field work.

(iii) At higher secondary school stage, conservation knowledge, problem identification and action skills are highlighted. Emphasis is given on teaching, practicals and field work.

(iv) At college stage, maximum stress should be on knowledge regarding sustainable development. Students should be taught about conservation, real life situations and awareness contents should include science based teachings, practicals and field work.

(v) At university stage, stress is given on teaching research and extension. Some research institutions, engineering colleges and IIT's

offer courses in environmental studies. At university stage, there are four major areas of environmental education viz. environmental engineering, conservation and management, environmental health and social ecology.

2. **Non-formal Environmental Education.** It deals with youths and adults from all segments of the society such as family, factory workers, managers, decision makers in environmental as well as non-environmental fields.

The non-formal environmental education is being imparted in our country in the following ways

(i) By celebrating the world environment day (i.e. on 5th of June).

(ii) By organizing eco-development camps for youth to create awareness about basic ecological problems.

(iii) By imparting in service and pre service training to the professionals, technical persons and legal experts.

32. **Assess National Green Cops (NGC).** [E] / [CO3]

NGC is a major initiative of MOEFCC for creating environmental awareness among young children. This scheme has indicated the importance of grass root level in enhancing the environment awareness at mass. Launched in 2001-02 the scheme aims at sensitizing the young minds towards environmental conservation and sustainable development. The scheme has come up with a network of more than 1, 00,000 Eco clubs across the country in 14 years, making it one of the largest conservation networks. This programme provides in-depth field experiences to school children and offers opportunities to convert their ideas into creative action.

33. **EvaluateTehri Dam Conflict.** [E] / [CO3, CO4]

The Tehri Dam Movement was an important environmental movement in India that opposed the construction of the Tehri Dam on the Bhagirathi River. The movement raised serious concerns about environmental safety, displacement of people, and the impact of large dams in the Himalayan region.

The Tehri Dam is located in the state of Uttarakhand. It was built on the Bhagirathi River, a major tributary of the Ganga. The dam project aimed to generate hydroelectric power, provide irrigation, and supply drinking water to several regions. However, the project also caused the submergence of the old town of Tehri and many surrounding villages.

The movement began in the 1970s and gained strength in the 1980s and 1990s. It was led by environmentalist Sunderlal Bahuguna, who strongly opposed the dam. He believed that building a large dam in the earthquake-prone Himalayan region was dangerous. He also argued that the project would lead to deforestation, loss of biodiversity, and displacement of thousands of people.

Many villagers protested against the dam because they were forced to leave their homes and agricultural lands. The rehabilitation process was criticized as inadequate. Activists organized peaceful demonstrations, hunger strikes, and awareness campaigns to stop or reconsider the project.

Despite strong protests, the construction of the Tehri Dam was

completed in the early 2000s. However, the movement played an important role in raising awareness about environmental protection, sustainable development, and the rights of displaced people.
In conclusion, the Tehri Dam Movement is a significant example of environmental activism in India. It highlighted the need to balance development with environmental safety and social justice. The movement remains an important chapter in India's environmental history.

34. **Appraise genetic diversity and species diversity.** [E] / [CO3]

GeneticDiversity
Genetic diversity refers to the variation in genes within a species. It includes differences in DNA among individuals of the same species or between populations.

- It enables organisms to adapt to changing environmental conditions.
- High genetic diversity increases survival and resistance to diseases.
- Example: Different varieties of rice or mango show genetic variation within the same species.

Species Diversity
Species diversity refers to the variety of different species present in a particular region. It includes both the number of species (species richness) and their relative abundance (species evenness).

- It contributes to ecosystem stability and productivity.
- High species diversity ensures better ecological balance.
- Example: Tropical rainforests have high species diversity compared to deserts

[5x5 = 25]

Part C

Essay

Answer any **2** questions

Each question carries 10 marks

35. Explain the components of environment.

All major components of the environment, namely, air, water, land, energy, and living beings including humans, are inter-related and inter-connected in many ways. Living beings are dependent upon their physical environment-land, water, air, and energy for their existence; the activities of the living organisms affect the physical components and the changed physical conditions again have a direct impact on living beings.

The components of environment are broadly classified as abiotic and biotic components. Abiotic

Components

Abiotic or nonliving components of an ecosystem include all the physical and chemical factors that influence living organisms.

•Examples of abiotic components are air, water, soil, rocks, etc.

•All basic inorganic elements and components such as soil, water, calcium, oxygen, and a variety of organic components, such as humus, and different climatic conditions such as light, temperature, humidity, wind, and precipitation, all are abiotic components of environment.

•Abiotic components are essential for the living world, as life cannot exist without factors such as sunlight, water, air, and minerals.

•The quality of the abiotic environment is a critical factor of life.

•Water, for example, can not only vary in terms of availability, but also in terms of characteristics such as turbidity, alkalinity, pH levels, conductivity, amount of dissolved oxygen, and the concentrations and combinations of various chemicals.

Biotic Components

•Biotic components are the living components of environment such as microbes, plants, animals, and human beings.

•They are classified as either producers or consumers, depending on the mode through which they obtain their food.

The biotic components of the environment are described as:

•Autotrophs or producers: These include green plants and algae which manufactures their own food.

•Green plants capture solar energy and, with the help of the chlorophyll pigment present in their leaves, make their own food in the form of sugars. This process in known as photosynthesis and the plants are known as photoautotrophs.

•Bacteria that utilize the oxidation of inorganic compounds such as hydrogen sulphide, ammonium, or ferrous ion as an energy source to manufacture their food by the process of chemosynthesis and ate known as chemoautotrophs.

Heterotrophs or consumers: The organisms of this group are not capable of manufacturing food and depend upon autotrophs to meet their food requirements.

•Animals, fungi, as well as most bacteria and protozoa depend on autotrophs for both energy and raw materials to make complex organic molecules.

•Heterotrophs are further classified as herbivores, carnivores, omnivores, and decomposers depending on their food habits

Decomposers: The non-green organisms like the fungi and some bacteria which are incapable of producing their food, live on the dead and decaying plant or animal parts are called decomposers.

36. **Discuss environmental education and its scope**. [U] / [CO1, CO2]

Population education has got a board scope with comprehensive subject matter as it is directly related with each and every human activity. Scope of population education varies according to the situation and need of the country. Consequently, its subject matters vary according to social, economic and political condition of country.

The scope of environment education is also called the content or subject matter of environment education. There are different aspects and components in the environment. Among them, the biological, physical, social and cultural aspects are important.

The scope of environmental education can be divided into biological, physical and sociological aspects. They are described below:

1. Biological Aspect: Biological aspects are one of the most important aspects of environmental education. Human being, animals, birds, insects, microorganism, plants are some of the examples of biological aspects.

2. Physical aspect: It can be further divided into natural aspects and human- made aspects.

Air, water, land, climate etc. are included in natural physical aspects. Likewise, Human made physical aspects cover all human made things such as roads, buildings, bridges, houses etc.

3. Socio- cultural aspect: Socio- cultural aspects are man-made social practices, rules and laws, and other religious places etc. Human beings have created them with their effort.

Environmental studies is very important and has following scopes:

1. There is a need for trained manpower at every level to deal with environmental issues like safe and clean drinking water, hygienic living conditions, clean and fresh air, fertile land, healthy food and sustainable development.
2. Since the pollution control laws becoming more stringent, industries need environmental experts to control pollution and disposal of wastes.
3. Environmental exports are now in great demand in industries for adopting green technologies to reduce pollution and to cut down the costs of effluent treatment.
4. These days, there is huge market world over for waste disposal and pollution control technologists and environmental experts.
5. If we want to live in a clean, healthy, aesthetically beautiful, safe and secure environment for a long time and wish to hand over a clean and safe earth to children, we will have to include environmental experts in all developmental on policy making committees.

37. **Evaluate Bishnoi movement.** [E] / [CO3, CO4]

The Bishnoi Movement is one of the earliest and most inspiring environmental movements in India. It is remembered for the great sacrifice made by the Bishnoi community to protect trees and wildlife. This movement took place in 1730 in Khejarli village near Jodhpur in the present-day state of Rajasthan.

The Bishnoi community was founded in the 15th century by Guru Jambheshwar, also known as Jambhoji. He preached 29 principles (Bishnoi means “29”), which emphasized living in harmony with nature, protecting trees, and showing kindness to animals. The Khejri tree was especially sacred to the Bishnoi people because it provided

food, fodder, and shade in the desert region.
In 1730, the ruler of Jodhpur, Maharaja Abhay Singh, ordered his soldiers to cut Khejri trees in Khejarli village for constructing his palace. When the soldiers arrived, a brave Bishnoi woman named Amrita Devi Bishnoi protested against the cutting of trees. She hugged a tree and declared that she would rather sacrifice her life than allow the tree to be cut. The soldiers killed her along with her three daughters.
Inspired by her courage, other villagers followed her example. In total, about 363 Bishnoi men, women, and children sacrificed their lives to protect the trees. Their sacrifice deeply moved the king, who later ordered a ban on cutting trees and hunting animals in Bishnoi villages.
The Bishnoi Movement is considered one of the first environmental movements in history. It highlighted the importance of environmental conservation long before modern environmental awareness began. The movement later inspired similar movements like the Chipko Movement.
In conclusion, the Bishnoi Movement teaches us the value of protecting nature and living in harmony with the environment. It remains a powerful symbol of courage, sacrifice, and environmental responsibility.

38. **Evaluate the objectives of IUCN Red lists and its various categories**. [E] / [CO3, CO4, CO5]

The IUCN Red List is a global inventory that evaluates the conservation status of plant and animal species. Its key objectives include:

1. **Assess extinction risk**
    - To scientifically evaluate the risk of extinction of species using standardized criteria.
2. **Provide a global database**
    - To create a comprehensive and updated list of species and their conservation status worldwide.
3. **Guide conservation planning**
    - Helps governments, NGOs, and researchers prioritize species and habitats for protection.
4. **Support policy and decision-making**
    - Provides data for environmental laws, biodiversity strategies, and international agreements.
5. **Raise public awareness**
    - Highlights endangered species and the need for biodiversity conservation.
6. **Monitor biodiversity trends**
    - Tracks changes in species populations over time to identify declines or recoveries.
7. **Encourage research and funding**
    - Identifies knowledge gaps and promotes scientific studies and conservation investments.

Categories of the IUCN Red List

Species are classified into the following categories based on their risk of extinction:

1. **Extinct (EX)**
    - No known individuals remaining.
2. **Extinct in the Wild (EW)**
    - Survives only in captivity or outside its natural habitat.
3. **Critically Endangered (CR)**
    - Extremely high risk of extinction in the immediate future.
4. **Endangered (EN)**
    - Very high risk of extinction in the near future.
5. **Vulnerable (VU)**
    - High risk of extinction in the medium term.
6. **Near Threatened (NT)**
    - Close to qualifying for a threatened category.
7. **Least Concern (LC)**
    - Low risk; species are widespread and abundant.
8. **Data Deficient (DD)**
    - Insufficient information to assess risk.
9. **Not Evaluated (NE)**
    - Species has not yet been assessed.

[10x2 = 20]